

COURSE DESCRIPTION			
Program	Fire Technology and Safety, Mechanical Engineering (Automobile Engg.), Mechatronics, Mechanical Engineering, Tool & Die Engineering, Wood and Paper Technology		
Course Title	Basic Mechanical Engineering		
Course Code	1021	Semester	I
Type of Course	Practicum	Contact Hours	90
Teaching Scheme (L: T:P:R)	3:0:3:0	Credits	4.5
CIE Marks	40	ESE Marks	100

Introduction

To get an idea about mechanical engineering and also to develop enthusiasm in mechanical engineering and related programmes.

Course Objectives

1	To introduce fundamental concepts of mechanical engineering
2	To provide basic knowledge of machines, manufacturing, and energy systems.
3	Familiarize students with power transmission systems, thermodynamics, and fluid power
4	Provide awareness of energy sources, maintenance, and industrial safety

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic Science			High School level

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Explain the scope of Mechanical Engineering, basic engineering concepts, units and basic principles of thermodynamics	Understand
CO2	Explain basic fluid mechanics, fluid power systems and power generation	Understand
CO3	Explain working principles of mechanical power transmission systems	Understand
CO4	Describe basic manufacturing and machining processes used in industry	Understand

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	-	-	1	1	-	-	1	-	2
CO2	2	2	2	1	2	2	-	-	1	-	2
CO3	2	2	2	1	2	1	-	-	1	1	2
CO4	2	2	2	2	2	2	1	-	1	1	2
Total	8	8	6	4	7	6	1	-	4	2	8
Correlation Level	2	2	2	1.33	1.75	1.5	1	-	1	1	2

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme					
							Theory			Practical		
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total
3	0	3	0	6	90	4.5	20	60	80	20	40	60

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	P
Module I	Theory	Scope and importance of Mechanical Engineering, Role of mechanical engineer in industry and society, Branches of Mechanical Engineering, Thermodynamic systems, processes and properties. Concept of energy, work, and heat, Laws of thermodynamics (Zeroth, First, Second - basics), Heat engines and refrigerators, Introduction to steam boilers and turbines, Basics of internal combustion engines	10	0
Module I	Practical	Identification of tools, machines, and materials, Identification of IC engines, Identification of parts of IC engines,	0	6
Module II	Theory	Properties of fluids, Pressure and its measurement, Pascal's law and applications, Hydraulic machines: pumps and turbines (basic idea), Pneumatic systems: components and applications, Industrial applications of hydraulics and pneumatics, Conventional energy sources: coal, petroleum, natural gas, Renewable energy sources: solar, wind, hydro, biomass, Advantages and limitations of renewable energy, Energy conservation and management	12	0
Module II	Practical	Identification of pressure measuring instruments, Identification of different types of pumps, Identify the different parts of the pump, Identify different types of turbine, Identify the parts of turbine, identify the components of pneumatic systems. experiment on Energy conservation and audit	0	9
Module III	Theory	Types of motion and power transmission, Belt drives: types, advantages, applications, Rope drives and chain drives, Gear drives: spur, helical, bevel, worm gears, Bearings: types and applications, Lubrication: purpose and methods	10	0
Module III	Practical	Identification of different types of belt drives and specifications, identification of chain drive parts, different types of gears, identification of different types of bearings, different methods of lubrication.	0	13
Module IV	Theory	Casting: principle, types of casting, components of casting, Forming processes: forging, rolling, extrusion, drawing, Machining processes: turning, drilling, milling, grinding (basic idea), Joining processes: welding, brazing, soldering, Surface finishing processes, Workshop safety rules, Personal protective equipment (PPE), Accident causes and prevention, Importance of safety in industries	13	0
Module IV	Practical	Identification of casting equipment and parts, Identification of different parts of lathe, identification of welding equipment, soldering practice, identification of safety equipment and its functions.	0	11

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Scope and importance of Mechanical Engineering, Role of mechanical engineer in industry and society, Branches of Mechanical Engineering	Explain the concept of mechanical engineering	L	CO1	Understand	1
Thermodynamic systems and processes, types of systems, different processes processes- isothermal, isobaric, isochoric, isothermal, adiabatic and polytropic processes (definitions only). Intensive and Extensive properties with examples	Explain the basic concept of thermodynamics	L	CO1	Understand	3
Concept of energy, work, and heat, Laws of thermodynamics (Zeroth, First, Second laws) definitions only	Explain the basic thermodynamic laws	L	CO1	Understand	1
Heat engines and refrigerators, Introduction to steam boilers basic idea and definitions only	Explain the concept of refrigerator, steam boilers	L	CO1	Understand	1
Basics of internal combustion engines - types of internal combustion engines, two stroke engines, four stroke engines, Diesel Engines and Petrol Engines explanations with figures.	Explain the idea of IC engines	L	CO1	Understand	4
Identify IC Engine, Refrigerator and different types of dismantling tools	Recognize heat engines, refrigerator and various tools and mechanisms used in the lab	P	CO1	Understand	3
Identify the various components of an IC Engine	Explain the basic parts of an internal combustion engine (Diesel/Petrol)	P	CO1	Understand	3
Module II					

Properties of fluids,density, specific gravity, specific weight,specific volume, surface tension- definitions with units	Explain the basic idea of properties of fluids	L	CO2	Understand	1
Pressure and its measurement- different types of pressure measuring instruments, manometer- types , piezometer, pressure gauges- basic principle only, Pascal's Law (definition only)	Explain the basic concept of pressure and its measurement	L	CO2	Understand	2
Hydraulic machines: pumps ,definition of pump, different types of pumps, centrifugal pump and reciprocating pump - basic working principle only	Explain the basic concept of pump	L	CO2	Understand	2
hydraulic turbines-definitions, types of turbines, Peloton turbine, Francis turbine and Kaplan turbine - basic working principle only	Explain the basic concept of turbine	L	CO2	Understand	2
Pneumatic Systems-definition, uses of compressed air, applications of pneumatic systems	Explain the basic concept of pneumatics	L	CO2	Understand	1
Conventional energy sources: coal, petroleum, natural gas- explanation only	Explain the idea of conventional energy sources	L	CO2	Understand	1
Renewable energy sources: solar, wind, hydro, biomass,Advantages and limitations of renewable energy- explanation only	Explain the idea of Renewable energy sources:	L	CO2	Understand	2
Energy conservation and management- importance of energy conservation,	Explain the Concept of energy conservation	L	CO2	Understand	1
Identify different types of pressure measuring instruments	Explain the pressure measuring instruments	P	CO2	Understand	3
Identify different types of hydraulic pumps	Explain hydraulic pumps	P	CO2	Understand	2
Identify different types of hydraulic turbines	Explain hydraulic turbines	P	CO2	Understand	2
Identify different types of pneumatic systems	Explain pneumatic components	P	CO2	Understand	2
	Series Lab Exam - I	P	CO2	Understand	3
Module III					
Types of motion and power transmission- different types of power transmission	Explain the types of motion and	L	CO3	Understand	1
Belt drives: types, -open belt drive, cross belt drive and V belt. Explanations with figure, materials for belt drive, advantages, applications	Explain the concept of belt drives	L	CO3	Understand	2
Rope drives and chain drives- explain with figure	Explain rope drive and chain drive	L	CO3	Understand	1
Gear drives: spur, helical, bevel and worm gears- explain with figure	Explain the concept of gear	L	CO3	Understand	2
Bearings: types and applications- roller bearing and journal bearing. List the types of roller bearings- application	Explain the concept of bearings	L	CO3	Understand	3
Lubrication:-principle of lubrication, different types of lubricants purpose and methods	Explain about lubricants and its application	L	CO3	Understand	1
Identify power transmission systems	Explain the different power transmission systems	P	CO3	Understand	3
Identify belt drive and chain drives	Explain the working of belt and chain drives	P	CO3	Understand	3
Identify different types gear drives	Explain various gear drive mechanisms and its working	P	CO3	Understand	3
Identify the different types of bearings used in industries	Explain the types of bearings	P	CO3	Understand	2
Identify different types lubricants and lubrication methods	Familiarize different types of lubricants and different methods of lubrication	P	CO3	Understand	2
Module IV					
Casting: principle, types of casting-basic concept only	Explain casting and its types	L	CO4	Understand	2

Components of casting-list the components	Explain the different types of moulding material	L	CO4	Remember	2
Forming processes: forging, rolling, extrusion, drawing-simple definitions only	Explain different types of forming processes	L	CO4	Understand	2
Machining processes: turning, drilling, milling, grinding (basic idea),	Explain different machining processes	L	CO4	Understand	2
Surface finishing processes-different types definition only	Explain different types of surface finishing processes	L	CO4	Understand	1
Importance of Safety, Workshop safety rules	Explain the importance of safety	L	CO4	Understand	1
,Personal protective equipment (PPE),Accident causes and prevention	Explain the importance of safety	L	CO4	Understand	2
Importance of safety in industries	Explain the importance of industrial safety	L	CO4	Understand	1
Identify various moulding and casting components	Explain and study the different components used in moulding/casting	P	CO4	Understand	3
Identify different types of metal forming process	Explain different types of metal forming process	P	CO4	Understand	3
Identify different types of machining process	Explain the different types of machining methods	P	CO4	Understand	3
Identify different surface finishing techniques	Explain different surface finishing methods	P	CO4	Understand	2
	Series Lab Exam - II	P	CO4	Understand	3

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Evolution and historical development of Mechanical Engineering,Contribution of Mechanical Engineering to sustainable development,Interdisciplinary nature of Mechanical Engineering (Mechatronics, AI, Robotics),State, path, and cycle concepts	5
2	Practical examples of energy conversion in machines,Units and measurement of work, heat, and energy,Mechanical work and flow work,Practical applications of Zeroth law,Practical limitations of energy conversion,Safety aspects of boilers ,Applications in power plants	5
3	Applications of IC engines in automobiles and power generation,Engine performance parameters (power, efficiency),Prepare comparison tables (SI vs CI engines, boilers vs turbines)	5
4	Watch videos/pictures/animations of various IC Engine components and different types of tools used in an automobile workshop	5
Module II		
5	Practical importance of fluid properties in engineering applications,Industrial applications of pressure measurement,Applications of Pascal's law;,Safety considerations in hydraulic systems	5
6	Applications of hydraulic pumps,Applications of hydraulic turbines,Components of pneumatic systems,Advantages of pneumatic systems,Applications in automation and manufacturing,Comparison between hydraulic and pneumatic systems	8
7	Environmental impact of coal usage,Formation and refining of petroleum,Applications of petroleum products in transportation and industry,Properties and types of natural gas,Environmental benefits of renewable energy,Economic advantages and energy security,Scope of renewable energy in India	5
8	Identify and watch the working of a hydraulic machine/pump used for domestic purposes. Watch videos related to pneumatic systems and pressure measuring devices. Know the difference between hydraulic turbines/pumps using web resources.	6
Module III		
9	Concept of motion in machines,Methods of power transmission,Mechanical power transmission systems in industries,Advantages and limitations of belt drives,Industrial applications of belt drives,Applications of rope drives in lifting and power transmission	5
10	Construction of chain drive,Applications of chain drives,Comparison between belt drive and chain drive,Difference between spur and helical gears,Function and types of bevel gears,Applications of worm gear drives	5
11	Bearing materials,Applications of bearings in machines,Purpose and importance of lubrication,Applications of lubrication in machines	5
12	Closely observe the type of drive mechanism used in your 2 wheeler/bicycle. Monitor any one gear drive used in your home/domestic purpose and capture its video for study purpose. Identify an application of lubrication used in your home. Capture its image and make a short note based on this image	5
Module IV		

13	Advantages, limitations, and applications of casting, Principle and types of forging, Applications of rolled products, Applications of extrusion, Principle of wire and tube drawing, Comparison of welding, brazing, and soldering	9
14	Purpose of surface finishing, Applications of surface finishing, Fire safety and emergency procedures, Selection and maintenance of PPE, Safety training and awareness, Role of management and safety officers, Safety culture and continuous improvement	10.5
15	Watch animations/video/documentaries of various manufacturing/machining/finishing operations done commonly in an industry and make a short note on the topic	6.5
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Basics of Mechanical Engineering	J Benjamin	Pentex Publications
2	Basic Mechanical Engineering	Raghuwanshi & Choubey	Laxmi Publications
3	Basic Mechanical Engineering	V. P. Arora	Tata McGraw-Hill Education

REFERENCE			
Sl. No.	Title	Author	Publication
1	Engineering Thermodynamics (Basic Concepts)	P. K. Nag	Tata McGraw-Hill Education
2	Fluid Mechanics and Hydraulic Machines (Introductory Level)	R. K. Rajput	S. Chand & Company Ltd.
3	Workshop Technology (Vol. I & II)	B. S. Raghuwanshi	Dhanpat Rai & Co.
4	Mechanical Engineering - Principles and Applications	P. N. Rao	Tata McGraw-Hill Education

WEB RESOURCES		
Sl. No.	URL	Modules Covered
	NIL	

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Internal Combustion Engines Petrol/Diesel-Disassembled engines to identify the parts	Any two/four stroke petrol/diesel engine components	1 or more
2	Manometer, Pressure gauges, etc. Hydraulic Pumps/Turbines	U tube manometer, pressure gauges of all types and any type of hydraulic pump/turbines used in labs.	1 or more
3	Chain drives, gear drive mechanism, ball/roller bearings, etc..	Any type of chain drive mechanism, gear drive systems and bearings used in automobile industry	1 or more
4	Lathe machines, milling machines, foundry, welding etc..	Mentioned equipments with any specification	1 or more

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology - Practicum

Mode	Attendance	CA1	CE	CA2	CA3	CA4	CA5	ESE	
		Average of Self learning activities from each module	Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)	Lab Examination (internal)
Duration				3 hours	3 hours	2 hours	2 hours	2.5 hours	3 hours
Exam mark		10	50	40	40	50	50	60	40

Converted to				10	10	10	10		
Marks	5	5	10	10		10		60	40
Total	40							100	
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60